ENLACE Summer Research Program 2025

Workflows for Wildfire Management

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Explanation of Workflow

1. Exploratory Data Analysis (EDA)

- **Input:** 250710_bp3d_db_data.csv (the raw dataset).
- **Process:**
  - The dataset is explored to understand its variables, types, and contents.
  - The purpose is to get a sense of data distribution, spot potential issues, and identify which features might be important.
- **Output Notebook:** EDA.ipynb documents the exploratory analysis.

2. Data Cleaning

- **Input:** Same raw dataset (250710_bp3d_db_data.csv).
- **Process:**
  - Remove duplicates, outliers, and null values.
  - Prepare a cleaned version of the dataset suitable for modeling.
- **Output:** df_cleaned.csv (cleaned dataset).
- **Associated Notebook:** data_preprocessing.ipynb handles the data cleaning steps.

3. Sensitivity Analysis (SA) using Machine Learning and SHAP Analysis

- **Input:** df_cleaned.csv (the cleaned dataset).
- **Process:**
  1. **Model Training:**
     - A **Random Forest Regressor** is trained.
     - An **XGBoost model** is also trained.
  2. **SHAP Analysis:**
     - SHAP (SHapley Additive exPlanations) values are calculated for both models to determine the importance of each input variable in the three target variables: surface consumed, midstory consumed, and canopy consumed.

    ▪ SHAP summary plots are generated to visualize the impact of features on predictions.

3. **Important notes:**

    ▪ Some data cleaning also happens in these notebooks. Our team tried to understand how the SHAP values would change depending on how we cleaned the dataset. After a careful discussion with Leticia and Pedro, we cleaned the data set in the following manner:

        ▪ Eliminating side to start ignition, blackline width, wind height, firing technique, dash length.

        ▪ Eliminating NaN values.

- **Outputs:**
    - SA_RF_SHAP.ipynb for the Random Forest SHAP analysis.
    - SA_XGBoost_SHAP.ipynb for the XGBoost SHAP analysis.

The previously described process can be seen visually in Figure 1.

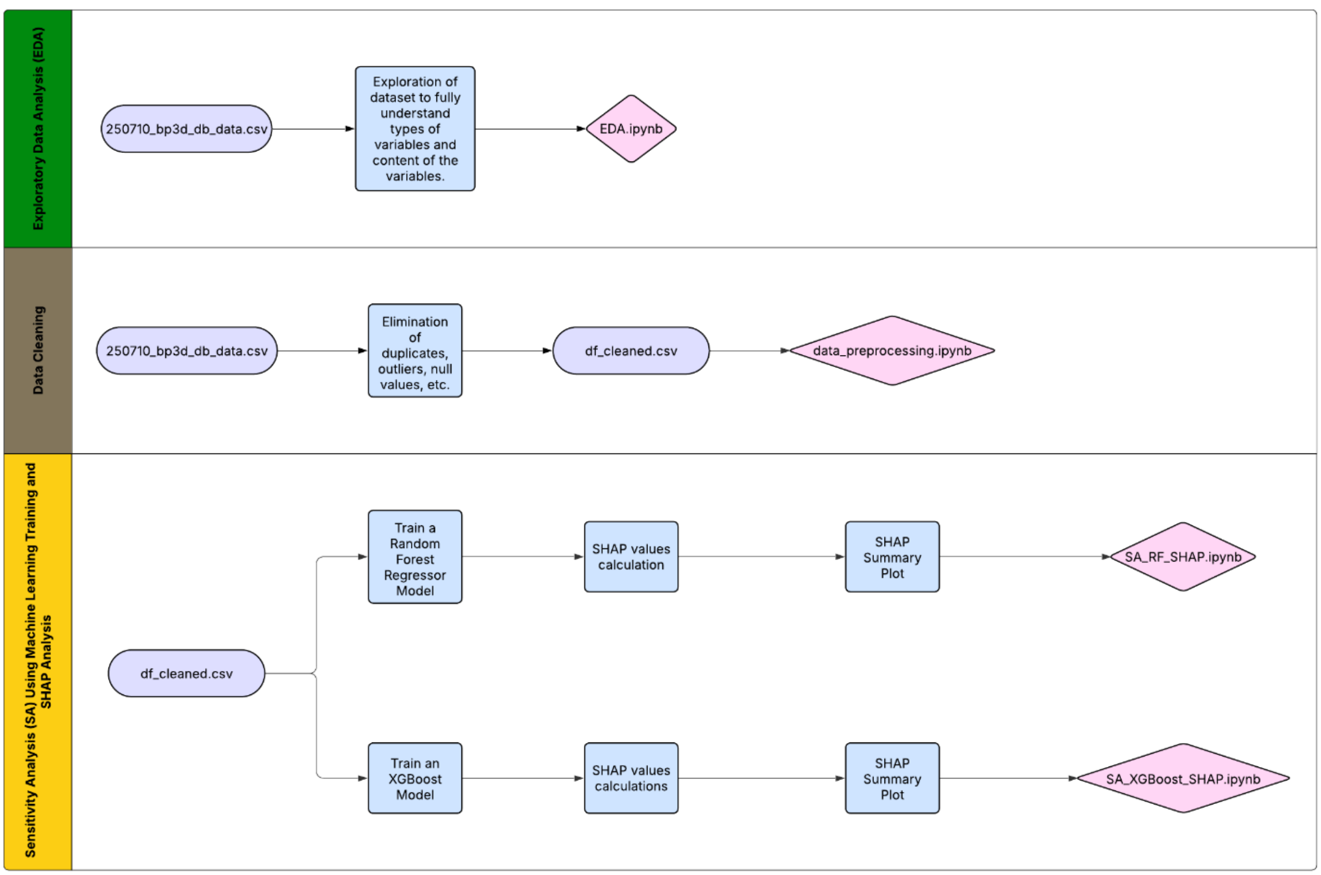

Figure 1